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Minimum Insert and Delete to Convert

Difficulty: **Hard**Accuracy: **27.85%**Submissions: **23K+**Points: **8**Average Time: **30m**

Given two arrays **a[]** and **b[]** of size **n** and **m** respectively, find the minimum number of **insertions** and **deletions** on the array **a[]**, required to make both the arrays **identical**.

Note: Array **b[]** is sorted and all its elements are distinct, operations can be performed at any index not necessarily at the end.

Examples :

Input: a[] = [1, 2, 5, 3, 1], b[] = [1, 3, 5]

Output: 4

Explanation:

Delete 2 from a: a[] = [1, 5, 3, 1]

Insert 3 after 1: a[] = [1, 3, 5, 3, 1]

Delete the last two elements: a[] = [1, 3, 5]

Total operations = 1 + 1 + 2 = 4.

Input: a[] = [1, 4], b[] = [1, 4]

Output : 0

Explanation: Both the Arrays are already identical.

Python3

Start Timer

```

1  from bisect import bisect_left
2
3  class Solution:
4      def minInsAndDel(self, a, b):
5          n, m = len(a), len(b)
6
7          # Store index of each element in b
8          pos = {value: idx for idx, value in enumerate(b)}
9
10         # Convert elements of a present in b to their indices
11         seq = []
12         for x in a:
13             if x in pos:
14                 seq.append(pos[x])
15
16         # Find LIS length
17         lis = []
18         for x in seq:
19             idx = bisect_left(lis, x)
20
21             if idx == len(lis):
22                 lis.append(x)
23             else:
24                 lis[idx] = x
25
26         lcs = len(lis)
27
28         # Minimum insertions + deletions
29         return (n - lcs) + (m - lcs)

```